

ENTERPRISE INFRASTRUCTURE PLAN

ABC Startup Solutions

IT Infrastructure Planning for a Newly Established Startup Company

Prepared for: Company Management / IT Manager

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Role: Junior System Administrator

ITEP 414 - System Administration and Maintenance

Week 2 Portfolio Project | Bachelor of Science in Information Technology

Laguna State Polytechnic University

PART 1. COMPANY PROFILE

Company Name

ABC Startup Solutions

Nature of Business

ABC Startup Solutions is a software development company that provides software development and technology-related solutions. The company needs an IT infrastructure that can support software development, office productivity, communication, file management, data storage and secure access to company resources.

Company Vision

The goal is to become an innovative software development company that provides effective technology solutions while always improving its services and supporting the growth of its employees and clients.

Company Mission

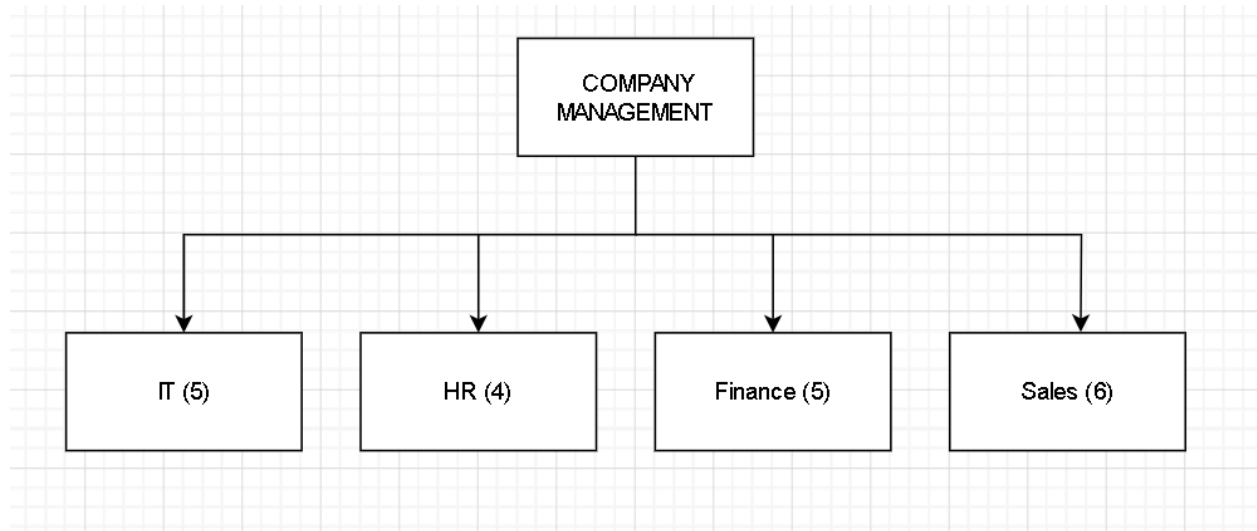
ABC Startup Solutions wants to provide quality software solutions through employees, reliable technology, efficient business processes and secure IT infrastructure. The company is dedicated to using technology to increase productivity, meet client needs and help the business grow in a way.

Office Location

The situation says that ABC Startup Solutions is on an office floor but it does not give a real location. For this infrastructure plan the company is considered to be working in a single-floor office environment.

Organizational Structure

The company is divided into four departments:



Employee Distribution

Department	Number of Employees
Information Technology	5
Human Resources	4
Finance	5
Sales	6
TOTAL	20

PART 2 — ENTERPRISE HARDWARE INVENTORY

The project requires desktop computers, laptops, a server, router, switch, printer, UPS, wireless access point, NAS storage, external backup drive, and monitors.

Hardware Inventory

Asset ID	Hardware	Quantity	Department	Purpose
HW-001	Desktop Computer	20	All Departments	Main workstation for employees
HW-002	Laptop	5	IT	Software development, administration, troubleshooting, and mobility
HW-003	Server	1	IT	Centralized services, files, applications, and internal resources
HW-004	Router	1	IT	Routes traffic between the internal network and the internet
HW-005	Managed Network Switch	1	IT	Connects computers, server, printer, and other wired devices
HW-006	Network Printer	2	HR/Finance	Printing business documents and records
HW-007	UPS	4	IT/Network Areas	Provides backup power and protects equipment from power interruptions
HW-008	Wireless Access Point	3	Office	Provides wireless network access throughout the office

HW-009	NAS Storage	1	IT	Centralized storage and backup
HW-010	External Backup Drive	2	IT	Offline backup and additional data protection
HW-011	Monitor	20	All Departments	Display for employee desktop workstations

Hardware Justification

Desktop Computers — 20

There are 20 employees, so each employee is assigned a desktop computer for daily work. This provides every employee with a dedicated workstation.

Laptops — 5

The five laptops are assigned to the IT department. IT employees may need mobile devices for troubleshooting, server administration, network configuration, testing, and other technical activities.

Server — 1

A server provides centralized services and resources for the company. Since the company is starting from zero infrastructure, a centralized server can support internal services and file management.

Router — 1

A router is required to connect the internal network to the internet and manage network traffic.

Switch — 1

The switch connects the company's wired devices, including employee computers, server, printers, and other network equipment.

Printers — 2

Two printers provide shared printing capabilities, particularly for departments such as HR and Finance that may handle documents and records.

UPS — 4

UPS devices protect critical IT equipment from sudden power interruptions. They can be assigned to the server, networking equipment, and other important equipment.

Wireless Access Points — 3

Three access points can provide wireless coverage throughout the single-floor office and help distribute wireless clients across the office.

NAS Storage — 1

The NAS provides centralized network storage and can be used as part of the company's backup strategy.

External Backup Drives — 2

Two external drives provide additional offline backup copies and help reduce dependence on a single storage device.

Monitors — 20

Each desktop workstation requires a monitor, so 20 monitors are provided for the 20 employees.

PART 3 — ENTERPRISE SOFTWARE INVENTORY

The PDF specifically requires the following software: Windows 11 Pro, Ubuntu Server, Microsoft Office, VS Code, Git, GitHub Desktop, VirtualBox, Google Chrome, Microsoft Defender, AnyDesk, and 7-Zip.

Software Inventory

Software	Version	License	Purpose
Windows 11 Pro	Current supported version	Commercial	Operating system for employee computers
Ubuntu Server	Current LTS version	Open Source	Operating system for the server
Microsoft Office	Current version	Commercial/Subscription	Word processing, spreadsheets, presentations, and office productivity
Visual Studio Code	Current version	Free	Software development and code editing
Git	Current version	Open Source	Version control for software projects
GitHub Desktop	Current version	Free	Graphical interface for Git and GitHub

VirtualBox	Current version	Open Source/Free	Virtual machine creation and testing
Google Chrome	Current version	Free	Web browsing and web-based applications
Microsoft Defender	Current version	Included with Windows	Malware and security protection
AnyDesk	Current version	Commercial/Free depending on use	Remote technical support and administration
7-Zip	Current version	Open Source	File compression and extraction

Software Explanation

Windows 11 Pro

Windows 11 Pro will be the operating system used on employee workstations. It gives the environment that employees need for office applications and company tasks.

Ubuntu Server

Ubuntu Server will be the server operating system. It is good for running server-side services. Gives a stable Linux environment.

Microsoft Office

Microsoft Office offers productivity applications for documents, spreadsheets, presentations and other business activities.

Visual Studio Code

VS Code is helpful for the IT department because software developers need a code editor for building and changing software projects.

Git

Git helps developers keep track of changes to source code and manage versions of software projects.

GitHub Desktop

GitHub Desktop offers an interface for working with Git repositories and can make managing repositories easier for people who like a graphical interface.

VirtualBox

VirtualBox lets IT employees and developers create machines for testing operating systems, applications and configurations.

Google Chrome

Chrome gives web browsing and access to web-based services and applications.

Microsoft Defender

Microsoft Defender gives security and malware protection, for Windows computers.

AnyDesk

AnyDesk lets authorized IT staff members remotely help users and fix computer problems.

7-Zip

7-Zip is used for compressing and extracting files, which can be helpful when moving or storing groups of files.

PART 4 — ENTERPRISE NETWORK INVENTORY

The network inventory must include the ISP modem, router, firewall, switch, access point, patch panel, CAT6 cables, and RJ45 connectors.

Network Equipment Inventory

Network Equipment	Quantity	Purpose
ISP Modem	1	Provides the connection between the office and the internet service provider
Router	1	Routes traffic between networks
Firewall	1	Protects the internal network from unauthorized network traffic
Managed Switch	1	Connects wired devices within the local network
Wireless Access Point	3	Provides wireless network connectivity
Patch Panel	1	Organizes and manages network cable connections
CAT6 Ethernet Cable	As required	Provides wired network connections
RJ45 Connectors	As required	Used to terminate CAT6 Ethernet cables

Network Design Explanation

The network begins with the internet connection from the ISP modem. The modem connects to the router, which manages network traffic. The router connects to the firewall, which provides an additional security layer before traffic reaches the internal network.

The firewall connects to the managed switch. The switch then distributes the network connection to the server, printer, wireless access points, and departmental computers.

PART 5 — ENTERPRISE NETWORK DIAGRAM

The PDF requires the network diagram to include:

- Internet
- ISP Modem
- Router
- Firewall
- Switch
- Wireless Access Point
- Server
- Printer
- IT Department
- HR Department
- Finance Department
- Sales Department

It must also have proper networking symbols, logical connections, labels, a title, and student name, and must be exported as **PNG and PDF**.

Exact Diagram Structure

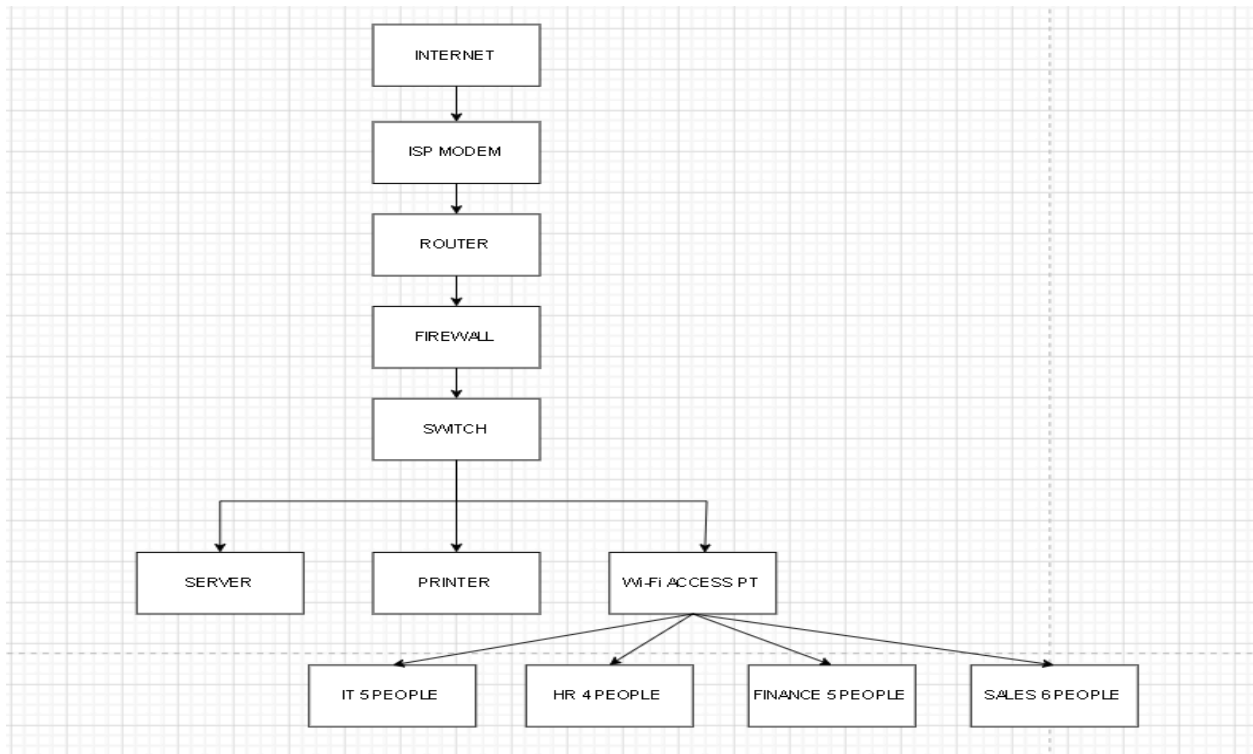


Diagram Title

ABC Startup Solutions — Enterprise Network Topology

Under the diagram

Prepared by: Lance Alday

PART 6. SYSTEM ADMINISTRATION ROLES

The assignment requires research on four positions. Asks for their responsibilities, skills, common tools and certifications followed by an explanation of how the four professionals work together.

Helpdesk Technician

Responsibilities

A Helpdesk Technician provides level technical support to employees. Common responsibilities include:

Troubleshooting computer problems

Assisting users with software issues

Installing and configuring software

Resolving basic network problems

Managing support requests

Documenting problems and solutions

Escalating complex issues, to other IT professionals

Skills

Important skills include:

Basic hardware troubleshooting

Operating system knowledge

Software troubleshooting

Basic networking

Communication

Problem-solving

Customer service

Documentation

Common Tools

Examples include:

Windows

Microsoft Defender

Remote support software

Ticketing systems

Command Prompt

PowerShell

Remote Desktop tools

Certifications

Possible certifications include:

CompTIA A+

CompTIA Network+

Microsoft-related certifications

Network Administrator

Responsibilities

A Network Administrator. Maintains the organization's network infrastructure.

Responsibilities include:

Configuring routers and switches
Maintaining network connectivity
Monitoring network performance
Managing wireless networks
Troubleshooting network problems
Implementing network security
Managing IP addresses
Maintaining network documentation

Skills

Important skills include:

TCP/IP
Routing and switching
Network troubleshooting
Network security
Wireless networking
Firewall configuration
Network monitoring
Documentation

Common Tools

Examples include:

Routers

Managed switches

Firewalls

Network monitoring software

Ping

Traceroute

Wireshark

Command-line tools

Certifications

Possible certifications include:

CompTIA Network+

Cisco CCNA

Other networking certifications

Linux System Administrator

Responsibilities

A Linux System Administrator manages Linux-based servers and services.

Responsibilities include:

Installing Linux operating systems

Configuring servers

Managing users and permissions

Installing software packages

Monitoring system performance

Managing storage

Maintaining server security

Troubleshooting Linux systems

Performing backups

Skills

skills include:

Linux command line

File permissions

User management

Networking

Shell scripting

System security

Server administration

Troubleshooting

Common Tools

Examples include:

Ubuntu Server

Bash

SSH

systemd

Linux command-line utilities

Package managers

Log management tools

Certifications

Possible certifications include:

Linux+

LPIC

Red Hat certifications

Cloud Administrator

Responsibilities

A Cloud Administrator manages cloud-based infrastructure and services.

Responsibilities include:

Managing cloud resources

Configuring virtual machines

Managing cloud storage

Managing user access

Monitoring cloud services

Implementing security controls

Managing backups

Monitoring resource usage

Supporting cloud-based applications

Skills

skills include:

Cloud computing

Virtualization

Networking

Identity and access management

Security

Storage management

Monitoring

Automation

Common Tools

Examples include:

Cloud management consoles

Virtual machines

Cloud storage

Identity and access management tools

Monitoring tools

Command-line cloud tools

Certifications

Possible certifications include:

AWS Certified Cloud Practitioner

Microsoft Azure certifications

Google Cloud certifications

How the Four Professionals Work Together

The four professionals work together to maintain an IT environment. The Helpdesk Technician is usually the point of contact when employees experience technical problems and can resolve basic issues or escalate them. The Network Administrator handles network-related problems involving connectivity, switches, routers, wireless access points and firewalls. The Linux System Administrator manages Linux-based servers. Ensures that server services, users, storage and security are properly maintained. The Cloud Administrator manages resources and services used by the organization. By working and communicating effectively these professionals can identify problems, assign them to the appropriate specialist, maintain infrastructure and ensure that employees can continue performing their work.

PART 7 — INFRASTRUCTURE RECOMMENDATIONS

The assignment specifically asks for recommendations concerning the internet provider, server specifications, backup strategy, security, antivirus, password policy, and expansion plan.

Internet Provider

ABC Startup Solutions should use a reliable business-grade internet service rather than relying on an entry-level residential connection.

The company has 20 employees across four departments, so internet connectivity will be important for software development, communication, online services, cloud applications, research, and business operations.

The selected provider should offer:

Reliable connectivity

Appropriate bandwidth for 20 employees

Business support

Stable service

A suitable service-level agreement where available

Server Specifications

A suitable initial server should have enough resources to support the company's current 20 employees while allowing some room for future growth.

Component	Recommendation
Processor	Multi-core server-grade processor
RAM	32 GB
Storage	SSD-based storage with additional capacity
Network	Gigabit Ethernet
Operating System	Ubuntu Server LTS
Backup	Connected to NAS and external backup system

UPS

Recommended

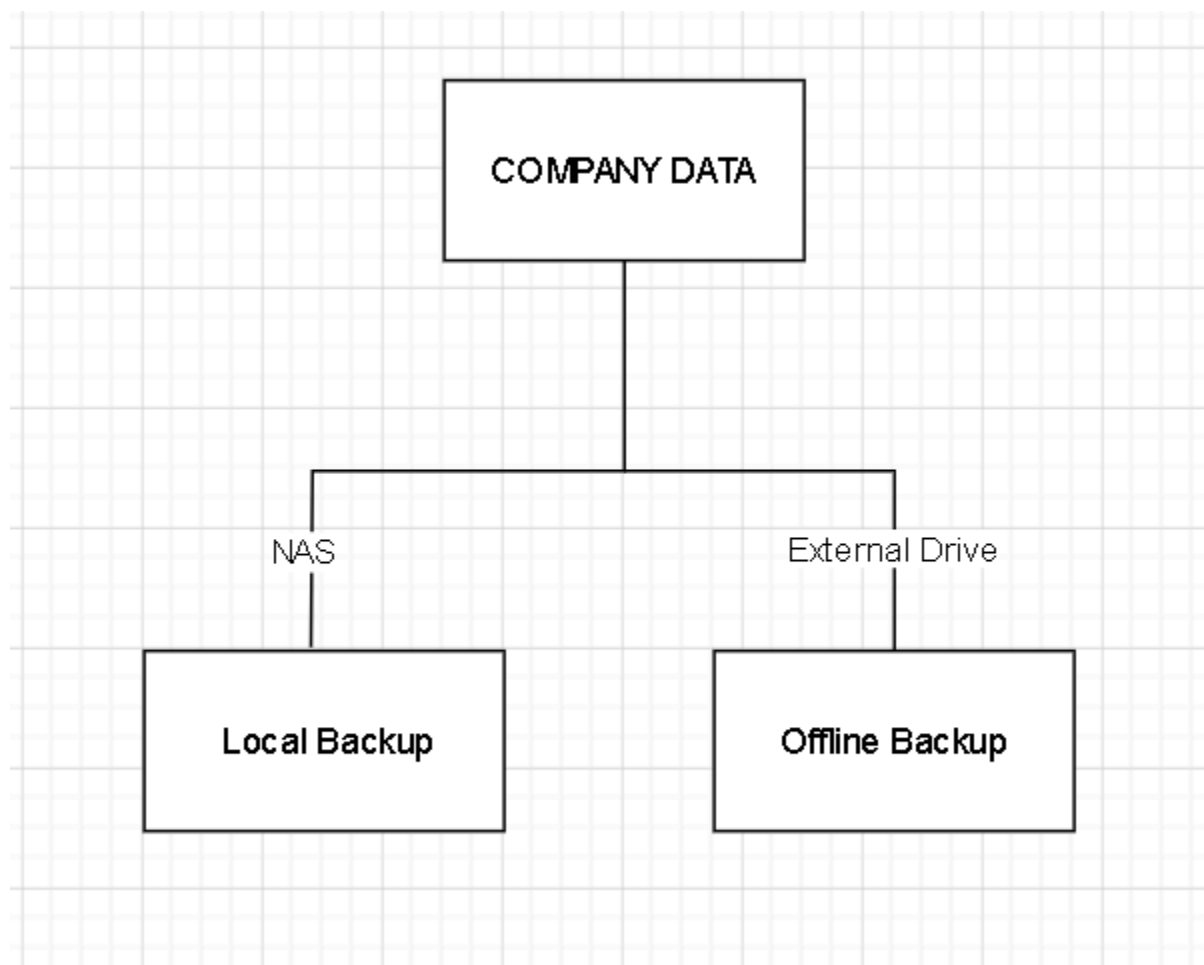
Purpose

Centralized services, storage, and internal resources

Backup Strategy

ABC Startup Solutions should use multiple backup locations rather than relying on only one copy of its data.

A practical strategy is:



The company should:

1. Store important data on the appropriate server/NAS storage.
2. Perform regular backups.
3. Maintain external backup copies.
4. Keep at least one backup copy disconnected when not actively being used.
5. Periodically test whether backups can actually be restored.

This reduces the risk of losing important business information because of hardware failure, accidental deletion, malware, or other incidents.

Security Recommendations

The company currently has no security policies, so security should be established before deployment.

Recommended measures include:

Firewall protection

Antivirus/endpoint protection

Strong passwords

Regular software updates

User access controls

Regular backups

Secure Wi-Fi configuration

Least-privilege access

Regular monitoring

Security awareness for employees

Proper documentation of IT systems

Antivirus

Windows computers should use Microsoft Defender as the baseline antivirus and security solution because it is one of the software requirements specified by the project.

Security software should be kept updated and configured to provide protection against malware and other common threats.

Password Policy

ABC Startup Solutions should establish a company-wide password policy.

Employees should:

Use strong and unique passwords.

Avoid sharing passwords.

Avoid using easily guessed information.

Avoid reusing passwords across important accounts.

Change compromised passwords immediately.

Use multi-factor authentication where available.

Lock or secure devices when leaving their workstations.

Administrative accounts should have separate credentials from normal user accounts.

Expansion Plan

The infrastructure should be designed so that ABC Startup Solutions can expand beyond its current 20 employees.

Future expansion can include:

Additional desktop computers

Additional laptops

Additional switches

Additional wireless access points

Increased server storage

Additional NAS capacity

Additional backup storage

Improved internet bandwidth

Additional network segments

Cloud services

Additional IT personnel

Planning for expansion prevents the company from having to completely redesign its infrastructure when the number of employees increases.

PERSONAL REFLECTION

Personal Reflection

This project helped me understand that system administration is not only about fixing computers or configuring networks. I learned that proper planning is an important part of building an IT infrastructure because a system administrator needs to understand the needs of an organization before purchasing equipment or installing services. In this project, I learned how to identify the hardware, software, and networking requirements of a company with different departments and responsibilities.

The most challenging task for me was designing the enterprise network because I needed to think about how the different components would connect with one another. The modem, router, firewall, switch, server, printer, wireless access points, and departments all need to be connected logically. Creating the inventory was also important because every piece of equipment needs a purpose and should support the company's operations.

Planning is important before deployment because it helps prevent unnecessary expenses, compatibility problems, security issues, and poor network performance. If a company purchases equipment without first determining what it actually needs, it may spend too much money on unnecessary devices or purchase equipment that cannot support its employees. Proper planning also makes future maintenance and expansion easier because the infrastructure is already documented and organized.

This project can help me become a better System Administrator because it gives me experience in thinking about an organization's IT requirements as a whole. Instead of looking at individual computers or devices, I learned how hardware, software, networking, security, storage, backups, and users are connected. I also learned the importance of technical documentation and professional presentation. The project showed me that a System Administrator needs technical knowledge as well as organization, communication, problem-solving, and critical-thinking skills. Overall, this activity gave me a better understanding of how IT infrastructure is planned before it is deployed and maintained.

